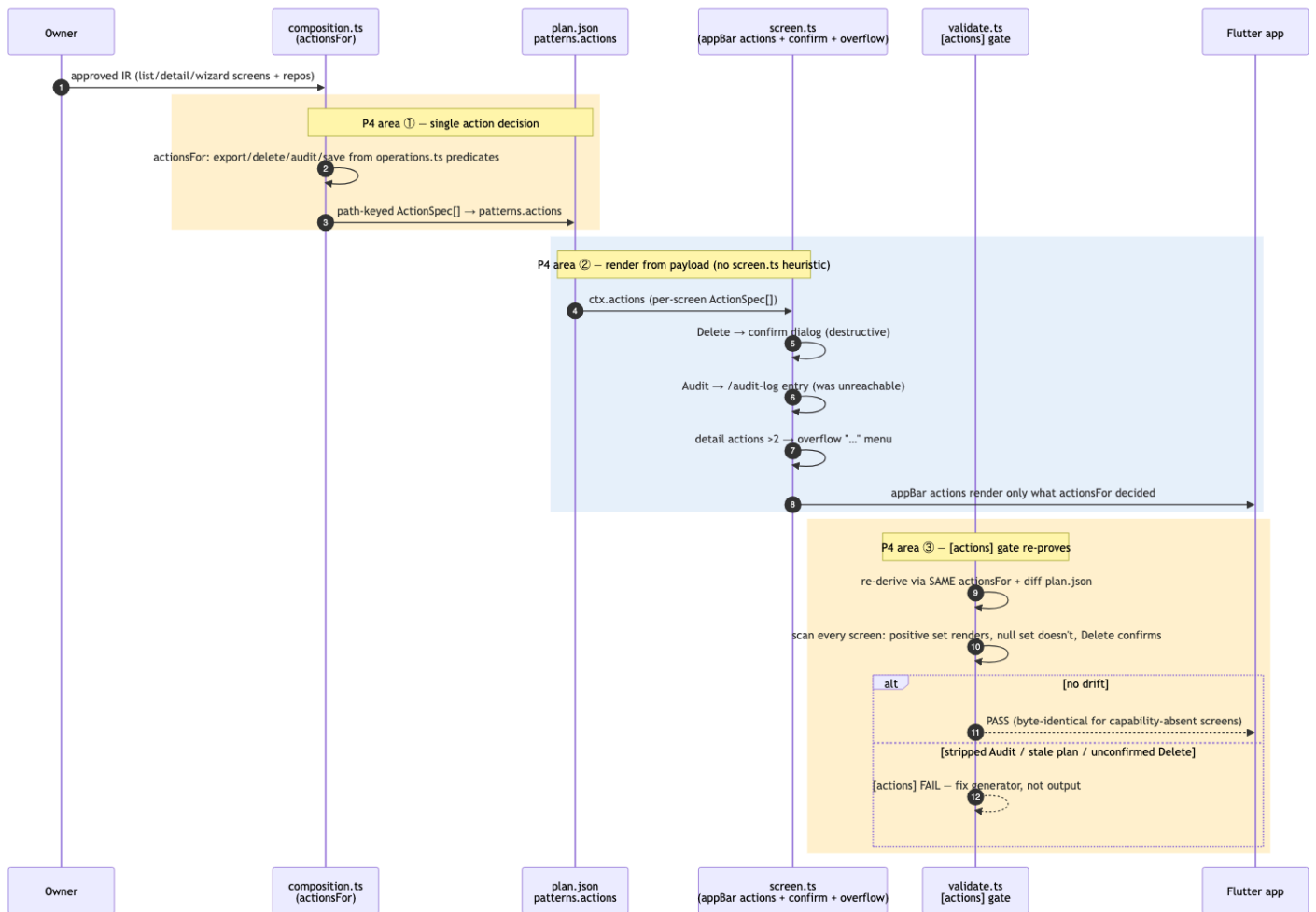
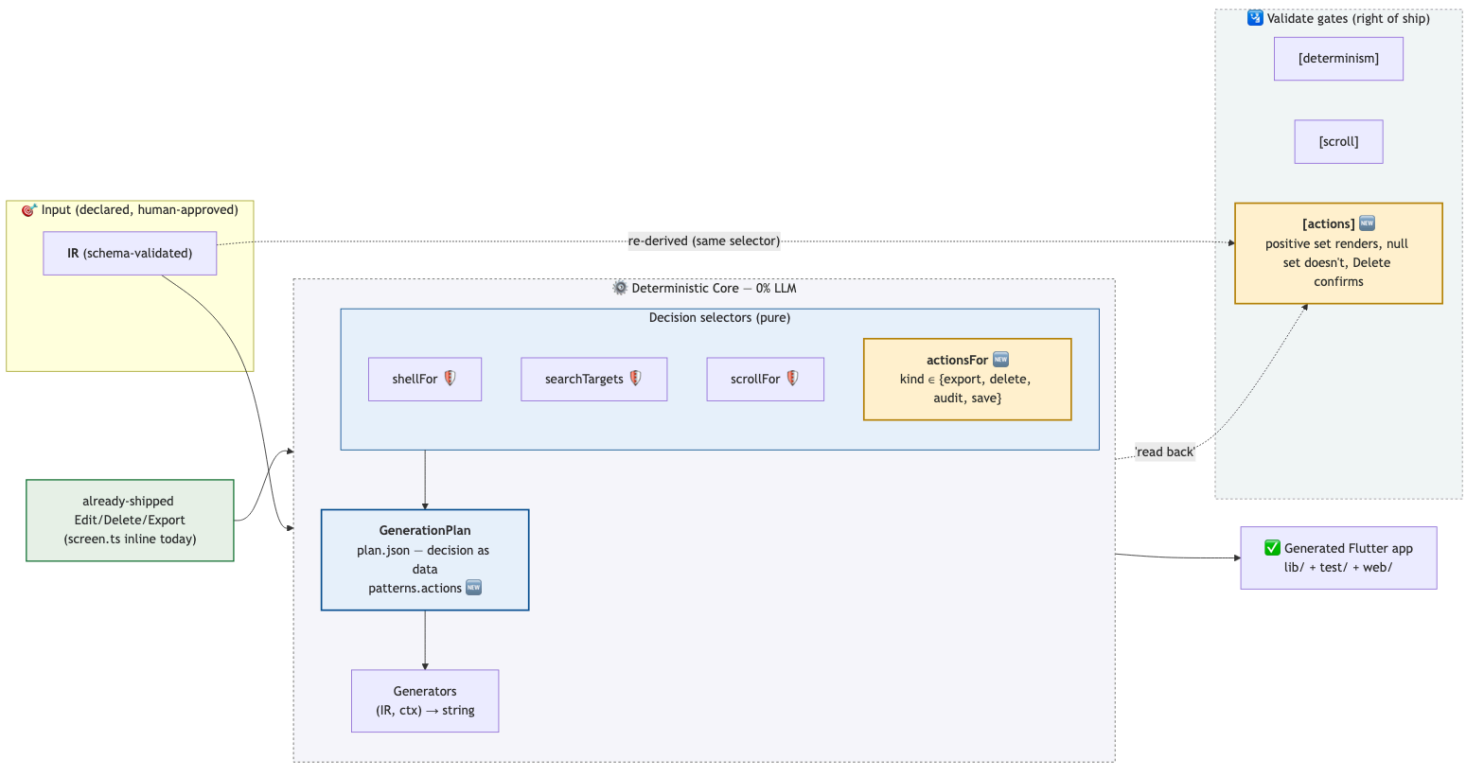


S-P4 — Capability-Driven Actions: Decision Brief > **Audience:** owner. **Purpose:** informed go/no-go on the P4 slice after the spike. > **Form:** executive summary → context → why → options (with priority/impact) → findings → > decision. Companion to `SPIKE\_PLAN.md` §P4, `INTERFACE\_PATTERN\_CONTRACT.md` §6, and > `research/SPIKE\_P4\_REPORT.md` (the full spike report, §17 format). **Status:** MODIFY — adopt reduced scope, subject to your go. **Date** 2026-08-17. --- ## 0. Diagrams — where the P4 action selector sits Legend: **amber** = what this brief decides (the `actionsFor` selector + `[actions]` gate); **light blue** = decision-as-data; **dashed** = gate groupings; **green** = already-shipped action emission being re-homed through the selector (consolidation, not new UI). ### Pipeline — actions decided once, rendered from the payload ### Sequence — one selector, Delete confirms, Audit becomes reachable ---



1. Executive summary

The spike inspected what the generator already ships vs what P4 proposes. **Edit, Delete, and Export** are **ALREADY capability-driven** with deterministic homes (`screen.ts` `appBarActions` + `resolveExport``) — P4's "derive from operations.ts" is partly done. What P4 genuinely adds, and is worth doing, is narrow: 1. **Delete confirm dialog** — today a delete is one tap, immediate, unrecoverable. 2. **Audit entry point** — ``AuditLogScreen` + `/audit-log` route` are *generated* (`hr_service`) but **unreachable from any button**. An audit trail you can't open is invisible. 3. **[actions]` gate + `patterns.actions`** — the decision-as-data hardening (same posture as `[scroll]/[search]/[plan-determinism]`). But **two parts of the proposal as written are rejected**: - **Save-as-extended-FAB** — the form already has an in-body `PrimaryButton('Create'/'Save')`. A FAB would render **two** save affordances on one form. Keep the existing button; `Save`` stays a documented kind only. - **A third action-decision home** — the spike found the risk of *fragmenting* one decision across `actionsFor`` + the existing inline blocks. It must be a **consolidation**, not a new parallel source of truth. **Decision: MODIFY**. **Reduced scope** = confirm dialog + audit entry + overflow "... menu + one selector + `[actions]`` gate, re-homing the existing Edit/Delete/Export emission. Estimated M (one slice). Byte-identical for screens with no applicable capability. ## 2. Context - Contract §6 (P4) + round-2 edit #3 froze the v1 `ActionSpec``: `{kind, label, icon, enablement, confirm?, presentation: inline|overflow|primary}`` — `params`` dropped. The 4-kind closed vocabulary (export/delete/audit/save) resolves grills C9 (schema), C10 (build-time vs runtime), C13 (icon allowlist), C16 (closed capability vocabulary). - P1/P2/P3 (shell/search/scroll) established the exact loop this slice reuses: selector in `composition.ts`` → plan.json payload → generator renders → validate gate re-derives. - Today's emission is inline in `screen.ts`` (`appBarActions` `:726-729``, `exportButtons` `:683-721``) — deterministic, but not represented in plan.json and not gated. P4 formalizes it. ## 3. Why it matters (impact) | Impact axis | With P4 (reduced) | Without | `---|---|---` | **Data safety** | Delete requires confirm (recoverable mistakes) | one-tap irreversible delete | **Audit value** | audit trail is reachable from the detail screen | generated-but-invisible audit (wasted capability) | **Trust boundary** | actions proven in plan.json + `[actions]`` gate | action decisions are unproven inline blocks | **Consistency** | ONE selector owns all action decisions | three inline homes (crudTarget/quickDecision/export) | **Goldens** | only screens gaining Audit/confirm change; capability-absent byte-identical | — | ## 4. Options (with priority + impact) | # | Option | Priority | Effort | Impact | Verdict | `---|---|---|---|---|---` | A | **Reduced P4**: `actionsFor`` single selector + Delete confirm + Audit entry + overflow "... + `[actions]`` gate, re-homing existing emission | P1 | M | High (data safety + audit reachability + trust hardening) | **RECOMMENDED (MODIFY)** | B | P4 as written (adds extended-FAB Save) | P1 | M+ | Negative on forms — two save affordances; more golden churn | rejected (regression) | C | Confirm dialog + Audit entry only, no selector/gate | P2 | S | Medium — fixes UX gaps but actions stay unrepresented in plan.json | acceptable fallback | D | Do nothing (status quo) | — | 0 | None — audit stays invisible, deletes unrecoverable, actions unproven | rejected | E | Extend scope: runtime per-user authorization around actions | P3 | L | High but needs the auth model (MF2 evolution), out of P4's build-time scope (C10) | DEFER (future auth spike) | ## 5. Findings (grounded, from code) - `operations.ts`` already provides every predicate P4 needs: `crudOperations`` (:39), `crudFormTargets`` (:73), `quickDecisionTargets`` (:101), `hasAudit`/isAudited`` (:381/:385), `resolveExport`/hasExport`` (:408/:431) — no new predicates required (C16 formalized). - Edit/Delete already inline on detail (`screen.ts:726-729``, decided by `canEditCreate`/canDelete`` at :246-247); Export already inline on list (:683-721, bloc-only guard :692). - **No `showDialog`/`AlertDialog` anywhere** in generated detail screens (delete is unconfirmed). - `hr_service`` `LeaveRequest`` is `audited:true``; `core/audit.dart`` + `core/audit_log_screen.dart`` + `/audit-log`` route all generated; `[audit]`` gate passes — but `rg`` finds **zero navigation to `/audit-log`**. The audit screen is unreachable. - Ledgerly is multi-feature (`features[]``), tasks/hr\_service/work\_auth single-feature with top-level `screens`` — `actionsFor``

must handle both (same flattening `shellFor` already does). - No `PopupMenuButton`/`more\_vert` anywhere today; overflow grouping is new. Only hr_service's LeaveRequest detail would reach >2 actions (Edit+Delete+Audit). - plan.json `patterns` per app: `{scroll, search}` (single-feature), `+ shell` (ledgerly). No `actions` key; `[actions]` gate is new. ## 6. Decision ****GO — Option A (MODIFY)**** Implement as one slice, per `SPIKE\_P4\_REPORT.md` §14: 1. `composition.ts`: `ActionSpec` (params` dropped) + `actionsFor(screen, entity, repo, ir)` — the SINGLE action decision + action→icon stem map. 2. `plan.ts`/`gen_context.ts`: `patterns.actions` (path-keyed) + `ctx.actions`. 3. `screen.ts`: re-home Edit/Delete/Export through `actionsFor`; wrap Delete in a confirm dialog; emit Audit action on audited details; overflow "... " when detail actions > 2. Byte-identical for capability-absent screens. 4. `crud_form.ts`: **\*\*no new FAB\*\*** — keep the in-body PrimaryButton (Save kind is type-only). 5. `validate.ts`: `[actions]` gate (re-derive via same `actionsFor`, diff plan, scan screens). 6. Verify: typecheck, 13/13 IRs, negative controls, CDP (confirm + audit + no overflow), byte-identical null-set proof. Then decision-brief PDF → Telegram. Acceptance: `[actions]` PASS on all 4 apps + all samples; prior gates still PASS; CDP confirms delete-confirm cancels/confirm and audit opens the log screen.